S.M. Zaker Abbasali[1] / M. Farsi[1] / M.R. Rahimpour[1]

Simulation and Dynamic Optimization of an Industrial Naphtha Thermal Cracking Furnace Based on Time Variant Feeding Policy

[1] Department of Chemical Engineering, School of Chemical and Petroleum Engineering, Shiraz University, Shiraz, Iran, E-mail: farsi@shirazu.ac.ir

Abstract:
The main goal of this research is modeling and simulation of ethylene production through naphtha thermal cracking in a coil furnace reactor. The naphtha cracking and decoking processes are modeled based on the mass, energy and momentum balance equations considering a detail kinetic model at pseudo-dynamic condition. To develop a reliable and applicable model, a detail thermal model is considered to predict tube skin temperature. To prove accuracy of developed model and considered assumptions, the simulation results are compared with the plant data. In addition, the results of coke burning in the decoking cycle by mixture of air and steam are presented. It is appeared that ethylene production decreases during the process run time gradually. The simulation results proved that increasing feed temperature, feed pressure and combustion chamber temperature during the process run time is a practical solution to maintain production capacity at an acceptable level. Finally, the optimal operating condition of thermal cracking process is calculated during the process run time based on the time variant feeding policy to achieve maximum ethylene production capacity. It is appeared that applying optimal condition on the system could increase ethylene production capacity from 15.31 ton.h^{-1} to 15.49 ton.h^{-1}.

Keywords: naphtha thermal cracking, decoking process, coil furnace, process modeling, dynamic optimization

DOI: 10.1515/cppm-2017-0032

Received: May 12, 2017; **Revised:** July 22, 2017; **Accepted:** August 14, 2017

1 Introduction

Ethylene is the most important feedstock because of low production cost and high purity. Generally, it is used to produce all types of polyethylene, ethylene dichloride, and ethylene oxide. Thermal cracking of hydrocarbons such as naphtha and ethane is the widest method to produce ethylene. Endothermic cracking reactions occur in the furnace coils at high temperature considering steam as dilution to decrease rate of coke formation [1]. Although, unsaturated hydrocarbons and aromatics are the main source of coke formation, the most of coke is produced by ethylene and propylene due to high mole fraction in the stream [2]. The produced coke is deposited on the coil surface and coke thickness increases continuously. Low thermal conductivity of coke decreases heat flux between coil and combustion chamber. Furthermore, increasing coke thickness increases pressure drop in the coil and leads to plant shutdown for decoking process. Since coke-burning is exothermic, steam is used as dilution to reduce over heating of coils. After that, in air polishing phase, only air is pass through the reactor to clean small amounts of coke left on the walls.

There are many studies on the thermal cracking of hydrocarbons to produce ethylene. The focused researches on the cracking could be classified on four main categories including cracking kinetic, decoking mechanism, process modeling, and process modification. Pant et al. investigated steam cracking of methyl cyclohexane in a tubular reactor [3]. Towfighi et al. developed a mechanistic reaction network model for cracking of LPG [4]. The parameters of developed kinetic model were calculated and used to simulate an industrial LPG cracking process. They also reported a kinetic model to predict produced coke on the coil tube surface. Ghassabzadeh et al. proposed an applicable semi-mechanistic kinetic model takes into account 172 reactions to predict yield distribution of products in the thermal cracking of kerosene [5]. Comparison between model results and experimental data proved the accuracy of the proposed kinetic model. Belohlav et al. developed a mathematical model to simulate the thermal cracking considering variety of feed types [6]. The combination of random

M. Farsi is the corresponding author.
© 2017 Walter de Gruyter GmbH, Berlin/Boston.

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

adaptive and relaxation algorithms was used to find the kinetic parameters. Gao et al. modeled a tubular reactor in a naphtha cracking furnace at steady state condition. The process condition was optimized considering profit as objective. The optimal temperature profile along the furnace, and steam-to-hydrocarbon ratio were calculated [7]. Ghashghaee and Karimzadeh focused on modelling and simulation of thermal cracking to predict dynamic characteristics of crackers during start-up [8]. Based on the developed models, fuel consumption, heat transfer rates and cracking temperatures during start-up were determined. Keyvanloo et al. proposed a model to predict performance of the naphtha thermal cracking process. They investigated the effect of temperature, steam-to-naphtha ratio, and residence time on the yield of light olefins [9, 10]. Barazandeh et al. used a one-dimensional model to analyze coil outlet temperature effect on the performance of cracking furnace [11]. Kumar et al. proposed a reaction network consists of first-order reactions for the thermal decomposition of naphtha in presence of steam. They used a jet-stirred reactor to investigate the effect of run time, surface area, reaction temperature, and inlet naphtha partial pressure on the rate of coke formation during naphtha cracking [12, 13]. Zheng et al. developed a three-dimensional model to predict distribution of heat flux and tube temperature considering a non-uniform distribution of surface heat transfer and the ethylene pyrolysis process [14]. Berreni et al. developed a one-dimensional pseudo-dynamic model to simulate and optimize thermal cracking of propane [15]. They investigated effect of gas temperature profile, and constant tube outer wall temperature profile on product yields and coking rate. Dynamic optimization showed that process profit can improve about 13.1%. Seifzadeh et al. modeled naphtha thermal cracking in presence of steam and carbon dioxide as dilution [16]. It was observed that using carbon dioxide as dilution can improve process run time about 25%. Cai et al. analyzed coke formation through steam cracking of ethane and naphtha [17]. They presented a good review about coke deposition. Zhang et al. developed a detail reaction model for decoking process. They coupled heat transfer and coke oxidation, the shrinking core model, and the zonal approach method in the model of plug flow reactor to predict coke thickness.

Since ethylene is one of the main products in petrochemical industry, even a small improvement in production capacity, may cause a great enhancement in production capacity and operating costs. In this research, a quasi-dynamic model is developed to predict performance of ethylene production plant. To create a reliable reference model, a reaction network consists of 30 reactions is considered. To prove accuracy of the developed model and assumptions, the simulation results are compared with the plant data. In addition, the optimal operating condition of thermal cracking process is calculated during the process run time based on the time variant feeding policy to achieve maximum ethylene production capacity.

2 Process description

Thermal and catalytic cracking are two main methods to crack heavy hydrocarbons to lighter ones such as ethylene and propylene. Thermal cracking has attracted more attention compared to the catalytic cracking to produce ethylene due to simplicity in design and operation. However, it suffers from some problems such as coke deposition on the surface, undesired product formation, and high energy consumption. The process scheme of thermal cracking is shown in Figure 1 (a).

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

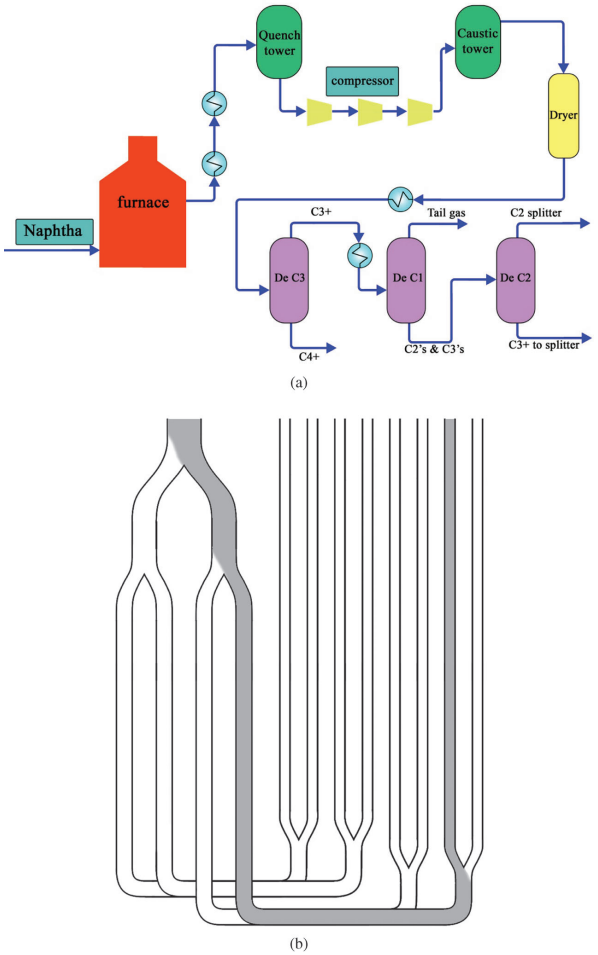


Figure 1: (a), Schematic of typical naphtha cracking process. (b), Schematic of coil furnace tube.

In the commercial process, steam and naphtha are mixed and preheated in the convection part of furnace to vaporize liquid feed. The reactions occur in coils, which have been placed in the radiation section. In this configuration, each two downward tubes are connected together at bottom of the furnace to form a larger upward tube. Then, each two upward tubes are connected together at top of the furnace to form a larger tube. Eighty side and bottom burners supply required energy for cracking reactions. Figure 1(b) shows that schematic diagram of coil furnace in Jam Petrochemical Company. The produced coke is deposited on the tube surface and increases heat transfer resistance in the system and causes a dramatic reduction in conversion and selectivity. In the considered plant, production run time and required time for decoking are 60 and 2 days, respectively. During the process run time a part of product is decomposed to coke. In the industrial plant, feed temperature, feed pressure and furnace temperature are increased gradually to overcome effect of coke deposition on the process conversion. In this plant, feed and combustion chamber temperature increase gradually from 888K to 905K, and 1423K to 1453K during 60 days. The composition of naphtha, and feed specifications are tabulated in Table 1 and Table 2.

Table 1: Inlet composition of feed stream.

Component	Specification
C_5+ liquid feed	
C_4H_{10} and lighter	max 1 wt%
C_5H_{12}	55–63 wt %
C_6 and heavier	36–44 wt%
Total sulphur max	530 wt ppm
C_3	1.7 wt%
Light ends liquid feed	
Iso-butane	9.6 wt%
Normal-butane	26.2 wt%
Iso-pentane	23.3 wt%
n-pentane	25.8 wt%

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

Iso-hexane	11.8 wt%
n-hexane	1.4 wt %
Cyclo-hexane	0.1 wt%
Benzene	0.1 wt%
Raffinate liquid feed	
Iso-butane	0.1 wt%
Normal-butane	0.9 wt%
Iso-pentane	12.3 wt%
n-pentane	6.2 wt%
Cyclo-pentane	0.3 wt%
Iso-hexane	28.7 wt %
n-hexane	10.1 wt%
Cyclo-hexane	3.5 wt %
Benzene	0.5 wt%
Iso- and n-heptane	29.2 wt%
Cyclo-heptane	0.6 wt%
Toluene	0 wt%
Iso- and n-octane	7.6 wt%
Cyclo-octane	0 wt%
C_8 aromatic	0 wt%
LPG liquid feed	
C_2–Cut	5.1 wt%x
C_3	31.3 wt%
Iso-butane	29 wt%
Normal-butane	34.1 wt %
Iso-pentane	0.5 wt%
C_5 cut liquid feed	
Butenes	1.2 wt%
Normal-butane	0.5 wt%
Iso-pentane	2.3 wt%
Iso-pentenes	17.6 wt%
n-pentane	31.9 wt%
n-pentene	14.5 wt%
Cyclo-pentane	25.8 wt%
Cyclo-pentene	2.9 wt%
Iso-hexane	0.5 wt%
Hexenes	1.9 wt%
n-hexane	0.2 wt%
Cyclo-hexane	0.1 wt%
Benzene	0.6 wt%

Table 2: Operating condition of process.

	Start of run	End of run
Feed flow rate (kg h^{-1})	44365	44365
Steam to feed ratio	0.5	0.5
Feed temperature (°C)	616	634
Furnace temperature (°C)	1150	1180
Feed Pressure (bar)	2.57	2.91

3 Mathematical modeling

3.1 Assumptions

To develop a simple and accurate mathematical model some assumptions have been considered as following:

- The feed stream is completely vaporized in the convection section of furnace
- No reaction takes place out of the cracking coils

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

– Produced coke is deposited on the inner surface of coil tube
– Plug flow regime due to high Reynolds number
– Ideal gas condition due to low pressure and high temperature
– Negligible dispersion in axial and radial direction due to high Peclet number

3.2 Mass and energy balance

Due to short residence time of feed stream in the coil and low rate of coke formation and deposition on the coil wall (one centimeter per sixty days), the steady-state material balance is acceptable. Thus:

$$\frac{dF_j}{dz} = \left(\sum_i s_{ij} r_i\right) \frac{\pi {d_t}^2}{4} \tag{1}$$

The energy balance can be written as:

$$\sum_j F_j c_{pj} \frac{dT}{dz} = Q(z)\,\pi d_t + \frac{\pi d_t^2}{4} \sum_i r_{ri} (-\Delta H)_i \tag{2}$$

The dynamic model of coke production is as:

$$\frac{d(D_t)}{dt} = \frac{F_c M w}{\rho_c \pi D_t} \tag{3}$$

Where ρ_c is specific gravity of coke. In the considered plant, specific gravity of coke is 1600 kg.m^{-3} [14]. Generally, there are five thermal resistances in the considered energy balance equation to connect chamber and gas temperature in the coil tubes. Figure 2 shows the considered thermal resistances in radial coordinate of coil tube.

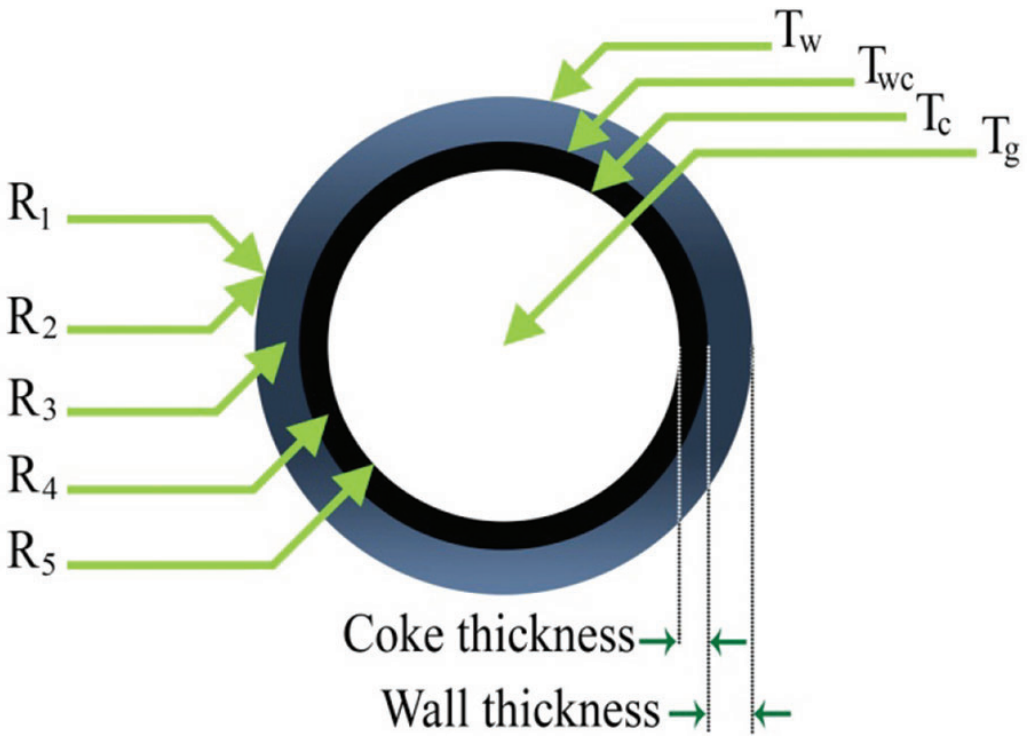


Figure 2: Considered thermal resistances in radial coordinate of coil tube.

Automatically generated rough PDF by ProofCheck from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

R_1, R_2, R_3, R_4 and R_5 are convection resistances in coil tube, conduction resistance of coke, conduction resistance of tube, radiation and convection resistances in combustion chamber, respectively.

$$R_{t1} = \frac{1}{2\pi r_i \Delta z h_g} \tag{4}$$

$$R_{t2} = \frac{\ln\left(\frac{r_w}{r_i}\right)}{2\pi k_C \Delta z} \tag{5}$$

$$R_{t3} = \frac{\ln\left(\frac{r_o}{r_w}\right)}{2\pi k_w \Delta z} \tag{6}$$

$$R_{t4} = \frac{1}{2\pi r_o \Delta z h_f} \tag{7}$$

$$R_{t5} = \frac{1}{\sigma \epsilon f_w \left(T_f + T_w\right)\left({T_f}^2 + {T_w}^2\right)(2\pi r_o \Delta z)} \tag{8}$$

$$U \times \mathrm{Ac} = \frac{1}{R_{t1} + R_{t2} + R_{r3} + \frac{1}{\frac{1}{R_{t4}} + \frac{1}{R_{t5}}}} \tag{9}$$

The rate of heat transfer between combustion chamber and process fluid in the coil is calculated by:

$$\mathrm{Q}\,(\mathrm{z}) = \mathrm{U} \times \mathrm{Ac} \times \left(T_{furnace} - T_g\right) \tag{10}$$

Where, k_w and k_c are thermal conductivity of tube and deposited coke on the surface. Conductivity of tube and coke are 5 $J.K^{-1}.m^{-2}$ and 5.5 $J.K^{-1}.m^{-2}$, respectively. To calculate heat of reaction, the following equation is used:

$$\Delta H_i = \left(\Delta H^0_{f\ at\ 298}\right) + \int_{298}^{T} C_{pi} dt \tag{11}$$

The pressure drop along the coil tube is calculated as:

$$-\frac{dP_t}{dz} = Fr\frac{G^2}{\rho_g} + G\frac{du}{dz} \tag{12}$$

Friction factor for straight tubes and bends are:

$$F_r = 0.092\frac{Re^{-0.2}}{d_i} \tag{13}$$

$$F_r = 0.092\frac{Re^{-0.2}}{d_i} + \frac{\xi}{\pi R_b} \tag{14}$$

Where:

$$\xi = \left(0.07 + .35\frac{\Lambda}{90^\circ}\right)\left(0.051 + 0.19\frac{d_t}{R_b}\right) \tag{15}$$

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

3.3 Kinetic models

In this research, a detail kinetic model consists of 28 reactions is considered to simulate the naphtha cracking process. In the considered model, olefins and aromatics are coke precursors. The reaction schmeme, kinetic model and rate parameters are selected form litrature [16]. Table 3 and Table 4 provide the kinetic, rate equation and rate parameters of considered model Table 5 presents the kinetic of decoking reactions.

Table 3: The kinetic and rate equation of considered model.

Reactions	Rate equations	No
Naphtha → $0.5H_2$+$0.76CH_4$+$1.16C_2H_4$ +$0.13C_2H_6$+$0.46C_3H_6$+$0.09C_3H_8$+$0.008C_4H_{10}$ +$0.185C_4H_8$+$0.113C_4H_6$+$0.08C_4H_4$	$r_1 = k_1\left[\frac{F_{C_6H_{14}}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[1]
$C_2H_6 \leftrightarrow C_2H_4+H_2$	$r_2 = k_2\left[\frac{F_{C_2H_6}}{F_t}\left(\frac{P_t}{RT}\right)\right] - k_{-2}\left[\frac{F_{C_2H_4}F_{H_2}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[2]
$C_3H_6 \leftrightarrow C_2H_2+CH_4$	$r_3 = k_3\left[\frac{F_{C_3H_6}}{F_t}\left(\frac{P_t}{RT}\right)\right] - k_{-3}\left[\frac{F_{CH_4}F_{C_2H_2}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[3]
$C_2H_2+C_2H_4 \rightarrow C_4H_6$	$r_4 = k_4\left[\frac{F_{C_2H_2}F_{C_2H_4}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[4]
$2C_2H_6 \rightarrow C_3H_8+CH_4$	$r_5 = k_5\left[\frac{F_{C_2H_6}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[5]
$C_2H_4+C_2H_6 \rightarrow C_3H_6+CH_4$	$r_6 = k_6\left[\frac{F_{C_2H_6}F_{C_2H_4}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[6]
$C_3H_8 \leftrightarrow C_3H_6+H_2$	$r_7 = k_7\left[\frac{F_{C_3H_8}}{F_t}\left(\frac{P_t}{RT}\right)\right] - k_{-7}\left[\frac{F_{C_3H_6}F_{H_2}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[7]
$C_3H_8 \rightarrow C_2H_4+CH_4$	$r_8 = k_8\left[\frac{F_{C_3H_8}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[8]
$C_3H_8+C_2H_4 \rightarrow C_2H_6+C_3H_6$	$r_9 = k_9\left[\frac{F_{C_3H_8}F_{C_2H_4}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[9]
$2C_3H_6 \rightarrow 3C_2H_4$	$r_{10} = k_{10}\left[\frac{F_{C_3H_6}F_{C_3H_6}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[10]
$C_3H_6+C_2H_6 \rightarrow C_4H_8+CH_4$	$r_{11} = k_{11}\left[\frac{F_{C_2H_6}F_{C_3H_6}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[11]
$C_4H_{10} \rightarrow C_3H_6+CH_4$	$r_{12} = k_{12}\left[\frac{F_{C_4H_{10}}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[12]
$C_4H_{10} \rightarrow 2C_2H_4+H_2$	$r_{13} = k_{13}\left[\frac{F_{C_4H_{10}}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[13]
$C_4H_{10} \rightarrow C_2H_4+C_2H_6$	$r_{14} = k_{14}\left[\frac{F_{C_4H_{10}}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[14]
$C_4H_{10} \leftrightarrow C_4H_8+H_2$	$r_{15} = k_{15}\left[\frac{F_{C_4H_{10}}}{F_t}\left(\frac{P_t}{RT}\right)\right] - k_{-15}\left[\frac{F_{H_2}F_{C_4H_8}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right] -kR_2\left[\frac{F_{C_4H_8}F_{H_2}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[15]
$C_4H_8 \rightarrow C_4H_6+H_2$	$r_{16} = k_{16}\left[\frac{F_{C_4H_8}}{F_t}\left(\frac{P_t}{RT}\right)\right]$	[16]
$C_2H_4+C_4H_6 \rightarrow C_6H_6+2H_2$	$r_{17} = k_{17}\left[\frac{F_{C_2H_4}F_{C_4H_6}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[17]
$C_3H_6+C_4H_6 \rightarrow C_7H_8+2H_2$	$r_{18} = k_{18}\left[\frac{F_{C_3H_6}F_{C_4H_6}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[18]
$C_4H_8+C_4H_6 \rightarrow C_8H_{10}+2H_2$	$r_{19} = k_{19}\left[\frac{F_{C_4H_8}F_{C_4H_6}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	[19]
$C_4H_6+C_4H_6 \rightarrow C_8H_8+2H_2$	$r_{20} = k_{20}\left[\frac{F_{C_4H_6}F_{C_4H_6}}{F_t^2}\left(\frac{P_t}{RT}\right)^2\right]$	(20)
$C_2H_4 \rightarrow$ Coke	$r_{21} = k_{21}\left[\left(\frac{F_{C_2H_4}P_t}{F_tRT}\right)^{1.34}\right]$	(21)
$C_3H_6 \rightarrow$ Coke	$r_{22} = k_{22}\left[\left(\frac{F_{C_3H_6}P_t}{F_tRT}\right)^{1.34}\right]$	(22)
$C_4H_8 \rightarrow$ Coke	$r_{23} = k_{23}\left[\left(\frac{F_{C_4H_6}P_t}{F_tRT}\right)^{1.34}\right]$	(23)
$C_6H_6 \rightarrow$ Coke	$r_{24} = k_{24}\left[\left(\frac{F_{C_6H_6}P_t}{F_tRT}\right)^{1.37}\right]$	(24)
$C_7H_8 \rightarrow$ Coke	$r_{25} = k_{25}\left[\left(\frac{F_{C_7H_8}P_t}{F_tRT}\right)^{1.37}\right]$	(25)
$C_8H_{10} \rightarrow$ Coke	$r_{26} = k_{26}\left[\left(\frac{F_{C_8H_{10}}P_t}{F_tRT}\right)^{1.37}\right]$	(26)
$C_8H_8 \rightarrow$ Coke	$r_{27} = k_{27}\left[\left(\frac{F_{C_8H_8}P_t}{F_tRT}\right)^{1.37}\right]$	(27)
Coke + $H_2O \rightarrow CO + H_2$	$r_{28} = k_{28}\left[\frac{F_{H_2O}P_t}{F_tRT}\right]$	(28)

Table 4: Arrhenius parameters for reaction.

Rate coefficient	A (s^1 or m^3 mol^1 $s^{1)}$)	E ($jmol^1$)
k_1	6.57×10^{11}	220141

Automatically generated rough PDF by ProofCheck from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

k_2	4.65×10^{13}	273021
k_3	7.28×10^{12}	273523
k_4	1.03×10^{12}	172747
k_5	3.75×10^{12}	273188
k_6	7.08×10^{10}	253008
k_7	5.89×10^{10}	214740
k_8	4.69×10^{10}	211852
k_9	2.54×10^{10}	247272
k_{10}	7.39×10^{12}	268667
k_{11}	1×10^{11}	251250
k_{12}	7×10^{12}	249700
k_{13}	7×10^{14}	295923
k_{14}	4.1×10^{12}	256692
k_{15}	1.64×10^{12}	261088
k_{16}	1×10^{10}	209340
k_{17}	8385000	144696
k_{18}	974000	149217
k_{19}	5.61×10^{11}	242708
k_{20}	1510000	124599
k_{21}	5×10^{14}	224000
k_{22}	2.77×10^{9}	116000
k_{23}	5.61×10^{18}	274000
k_{24}	1.39×10^{12}	141000
k_{25}	1.39×10^{12}	141000
k_{26}	1.39×10^{12}	141000
k_{27}	1.39×10^{12}	141000
k_{28}	50900	238000
k_{-2}	849000	849000
k_{-3}	381000	381000
k_{-7}	903	903
k_{-15}	17800	17800

Table 5: Decoking kinetic and rate reactions.

Reactions	Rate equations	No
$C_{(s)}+O_{2(g)} \rightarrow CO_{2(g)}$	$r_{30} = 270504exp\left(-\frac{188800}{RT}\sqrt{P_{O_2}}\right)$	(29)
$C_{(s)}+ H_2O_{(g)} \rightarrow CO_{(g)} + H_{2(g)}$	$r_{31} = 76387exp\left(-\frac{238260}{RT}P_{H_2O}\right)$	(30)

4 Optimization

Process optimization is defined as a systematic procedure to find the desired operating condition of a process to maximize or minimize an objective function considering operational, economic and environmental limitations. In the process optimization, after problem understanding and identification, selection of an objective and set of decision variables is the main step. Since the subject of cracking plant is large scale ethylene production, thus annual ethylene production capacity has selected as objective function. Generally, temperature policy is one of the main key to control the thermal cracking reactions. Thus, in this work, dynamic trajectories of feed, and furnace temperature, and optimal value of steam to feed ratio have selected as decision variables. The feed and furnace temperatures at initial and end of run time are calculated and assumed that temperature change linearly during the process run time. The next step in the problem formulation is determining constraints and bounds. Due to coil melting at high temperature and based on the plant restrictions, the maximum skin temperature is set to 1533K. In addition, to prevent racking in the feed preheater, the maximum feed temperature is set at 953K. In addition, the steam to feed ratio lies between 0.4 and 1. The developed mathematical model connects the decision variables and objective function together. After process modeling, selection of an optimization method is necessary. In this research, genetic algorithm as a global optimization method has selected as a powerful method to find the optimal condition of process [18, 19]. The population size and number of generations set to 200 and 200, respectively. The formulated optimization objective and constraint could be written as:

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

$$J = \max \int_0^{t_f} dF \tag{16}$$

$$0.4 < \frac{Steem}{Feed} < 1 \tag{17}$$

$$T_{Skin} < 1533\text{K} \tag{18}$$

$$T_{Feed} < 953\text{K} \tag{19}$$

4.1 Genetic algorithm

In computer science, genetic algorithm (GA) is a metaheuristic search method inducted by the evolutionary ideas of natural selection and genetic. The method starts from a population of randomly generated individuals in search domain. In each generation, the fitness of every individual in the population is evaluated, and a part of individuals are selected from the current population based on the fitness value, and some of individuals are modified to form a new generation by crossover and mutation. Crossover concerns two selected individuals as parents, that exchange parts of their genomes to form two new individuals as children. Mutation consists in flipping bits of individual's strings at random. The new population is then used in the next iteration of the algorithm to calculate optimal solution. Commonly, the algorithm terminates when either a maximum number of generations has been produced, or a satisfactory fitness level has been reached for the population.

5 Results and discussion

5.1 Model validation

In this section, the simulation results are compared with the plant data. Also, Table 6 presents a comparison between simulation results and plant data in Jam Petrochemical Company at same process condition. It was mentioned that the process run time is about 60 days. The appeared agreement between simulation results and plant data proves the accuracy of the developed model and assumptions.

Table 6: Comparison between modeling results and industrial data.

Component	Industrial data	Model result	Relative error
Ethylene (ton h^{-1})	15.23	15.31	0.52
Propylene (ton h^{-1})	6.85	6.62	3.34
Methane (ton h^{-1})	2.92	2.88	1.37
Hydrogen (ton h^{-1})	0.27	0.267	1.11
Ethane (ton h^{-1})	0.51	0.52	1.96
Mean Outlet Temperature (K)	1138.2	1150	1.16
Skin Temperature at SoR (K)	1323	1327	0.3
Skin Temperature at EoR (K)	1387	1387	0

5.2 Cracking process

In this section the calculated temperature, pressure and component molar flow rates are presented along the coil tube. Figure 3(a–b) shows ethylene and naphtha molar flow rates along the coil tube at different run times. It appeared that naphtha is cracked to product and decreases along the length of coil tube. Since produced coke is deposited over the coil tube surface, it increases pressure drop in the coil during the run time. Since rate

Automatically generated rough PDF by ProofCheck from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

equations are pressure dependent, increasing pressure drop has a negative effect on the reaction rates. Coke deposition on the surface decreases rate of naphtha cracking considerably, so ethylene production decreases from 155.5 mol.s^{-1} at plant start up to 151.9 mol.s^{-1} at the end of run time. Since naphtha cracking decreases during the process run time, it changes product distribution in the furnace effluent. Applying higher temperature in the furnace could control naphtha conversion at the required level. However, since the thermal cracking reactions are temperature and pressure dependent, applying higher temperature in the coil tube changes the product distribution in the plant. On the other hand, it increases rate of coke formation and deposition.

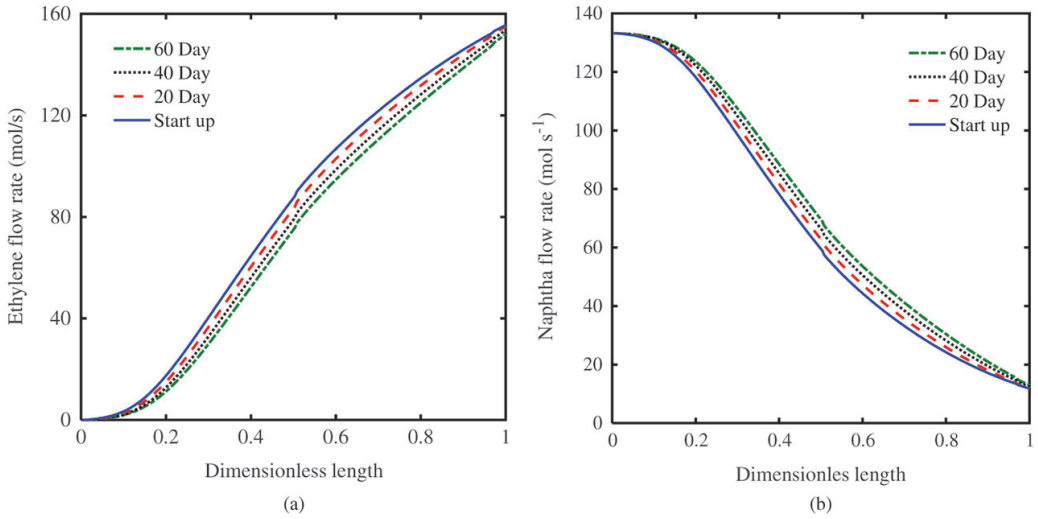


Figure 3: (a), Ethylene production rate along the coil tube. (b), Naphtha cracking rate along the coil tube.

Figure 4(a–e), shows the hydrogen, propylene, butylene, methane and ethane molar flow rates along the coil tube. It appeared from presented reaction network that hydrogen is produced through naphtha cracking reaction and consumed through other reactions. Thus, decreasing naphtha conversion during the process run time decreases rate of hydrogen production in the cracking process. According to Le Châtelier's principle, when an independent variable of an equilibrium system changes, the equilibrium shifts in the direction that tends to reduce the effect of the change. Thus, increasing hydrogen concentration along the coil tube decreases rate of ethylene and propylene dehydrogenation as reversible reactions. Other valuable side products in the naphtha-cracking units are propylene and butylene. The results show that propylene production decreases from 45.05 mol.s^{-1} at plant start up to 43.8 mol.s^{-1} at the end of run time. In addition, butylene experiences a uniform production level, about 20.1 mole s^{-1}, during process run time. Generally, production of heavy olefins reduces with increase in number of carbons, so butylene is in lowest concentration compared to the ethylene and propylene.

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

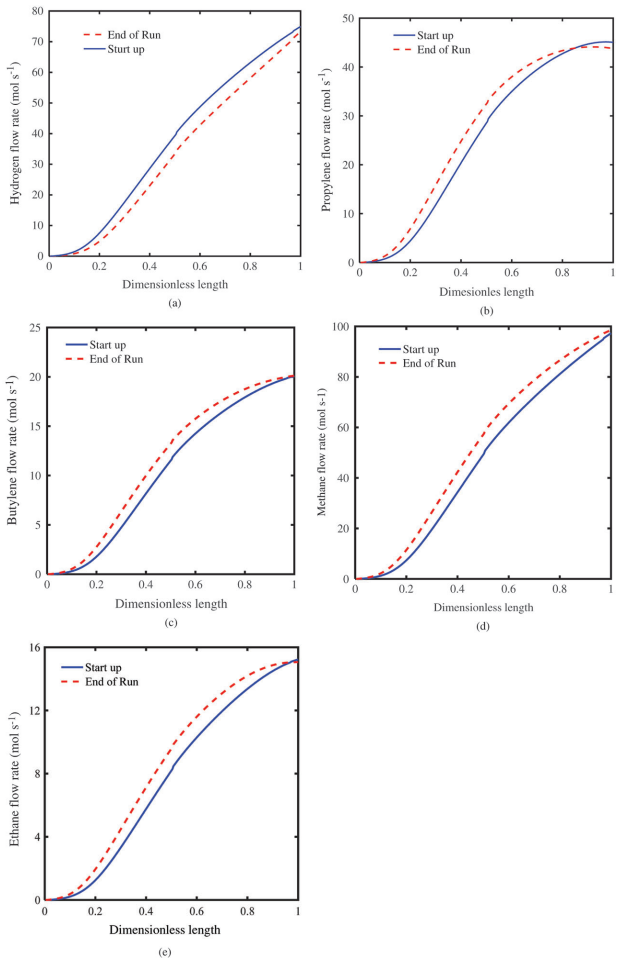


Figure 4: (a), Hydrogen molar flow along the coil tube. (b), Propylene molar flow along the coil tube. (c), Butylene molar flow along the coil tube. (d), Methane production rate along the coil tube. (e), Ethane molar flow along the coil tube.

Figure 5(a–c) shows gas, coil outlet (outlet product from furnace), and wall temperature profiles along the coil tube. Since radiation heat transfer depends on wall temperature, a trial and error method is used to calculate it. First a wall temprature is considered for each element along the coil tube length. Then, the wall temperature is calculated by heat balance considering inner (heat transfer between gas and wall) and outer heat fluxes (heat transfer between wall and chamber) as criteria. By compering inner and outter fluxes, wall temprature is corrected untill the fluxes get close to each other. Based on the simulation results the coil outlet temperature increases linearly from 1113K to 1133K during the process run time due to applied feed and combustion chamber temperature profiles. in addition, the maximum skin temperature is 1327K and 1387K at the plant startup and end of run, respectively. Increasing coke deposition on the coil tube surface increases heat transfer resistance between furnace chamber and coil tube. Indeed, applying higher furnace temperature to keep production capacity at an acceptable level and increasing thermal resistance due to coke formation are the main reason to increase skin temperature during the process run time. It was mentioned that, the temperature of combustion chamber increases from 1423K to 1453K linearly during the process run time to keep naphtha conversion in an acceptable level.

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

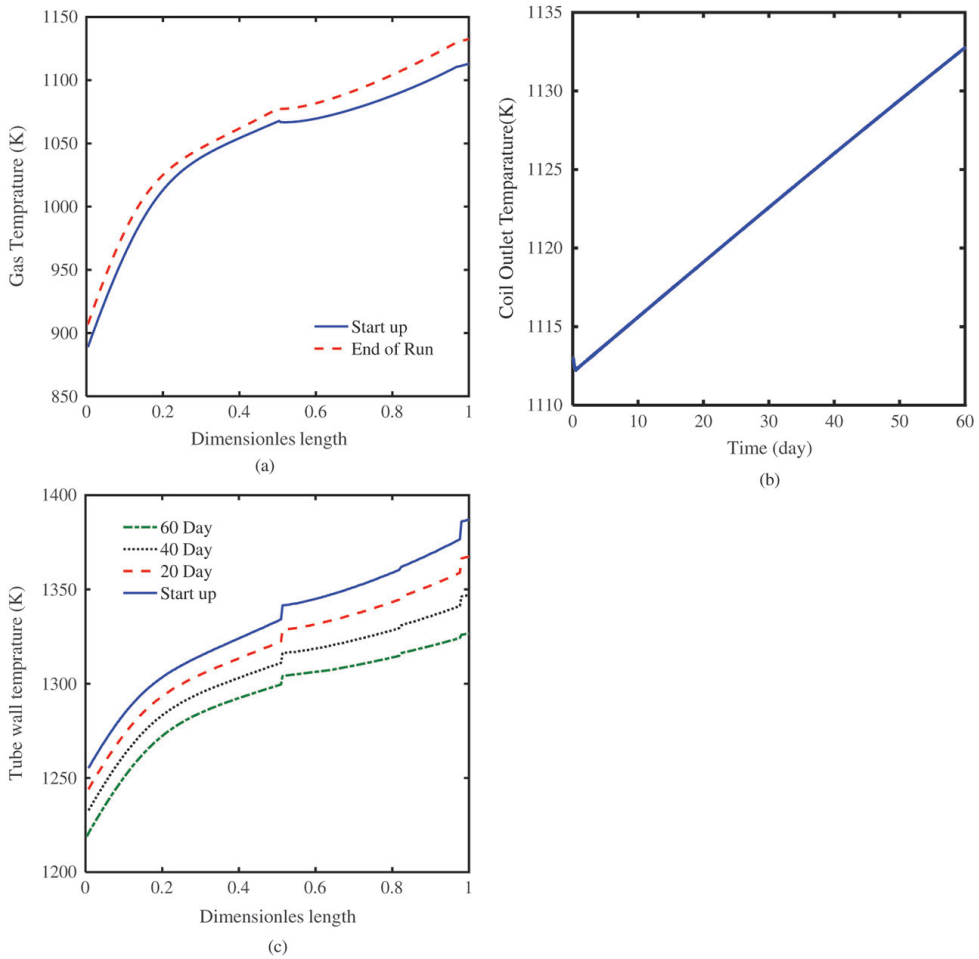


Figure 5: (a), Gas temperature along the coil tube length. (b), Coil outlet temperature during process run time. (c), Wall temperature along the coil tube length.

Figure 6(a-b) shows the inner diameter of the coil tube and pressure drop along the tube length versus run time. The inner diameter of the downward and upward tubes is 4.8 cm and 6.8 cm at clean condition, respectivelly. Coke deposition decreases effective tube diamater. Generally, rate of coke build up on the wall depends on gas phase temprature, concentration of coke precursors such as aromatics and olefines, and reactant residence time. Due to low concentration of coke precursors and temperature, coke deposition on the inner surface of downward coil tubes is sufficiently smaller than the upward parts. Generally, the temperature profile along the coil tube has a good agreement with the coke build up on the tube. Higher temperature results in higher coke deposition and lower effective diameter for flowing gas stream. Because of friction, bends in connection points and high gas velocity, pressure drops is about 0.5 bar at the start up. Increasing coke thickness in the coil tube increases the pressure drop to 0.9 bar at the end of run time. It appeared that pressure drop increases during the process run time due to coke deposition on the wall and decreasing the effective diameter of coil tube. In the real plant, since the outlet product from coil is fed to the next units, the feed preesure increases from 2.57 bar at the startup to 2.91 bar and the end of run to keep outlet presure in an acceptable range.

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

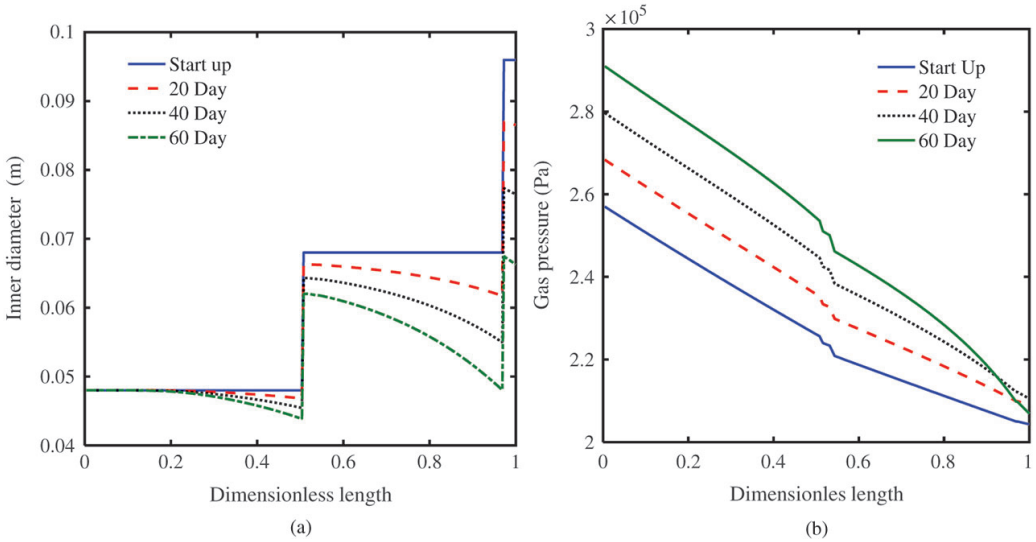


Figure 6: (a), Inner diameter of coil tube. (b), Pressure drop along the coil tube.

5.3 Decoking process

The formed coke is deposited on the inner surface of coil tubes and coke thickness increases continuously. Increasing the coke thickness leads to a plant shutdown for decoking process by mixture of air and steam. Generally, the decoking process contains depressurizing, cooling, coke burning, heating and pressurizing stages. The deposited coke is burned at coke burning process and other stages are preliminary. Since coke-burning reaction is exothermic, steam is used as dilution to reduce over heating of coils. Figure 7(a–b) shows the coke thickness and temperature profiles along the coil tube during the decoking process. It appeared that the deposited coke on the downward tubes is burned after 8 h, while about 24 h is required to burn the deposited coke on the surface of upward tubes. In the decoking step, oxygen reacts with the coke and coke molequles converted to carbon dioxide and water vapor. Thus, coke thickness decreases continously. It appeared that heat generation during coke combustion increases gas temperature along the coil tube. Inspite of naphtha cracking process that heat is transferred from combustion chamber toward the coil tube, the heat is transferred from coil tube toward the combustion chamber in the decocking process. Furthermore, since the deposited coke on the tube is burned and coil tube is cleaned versus time, temperature decreases with passage of time.

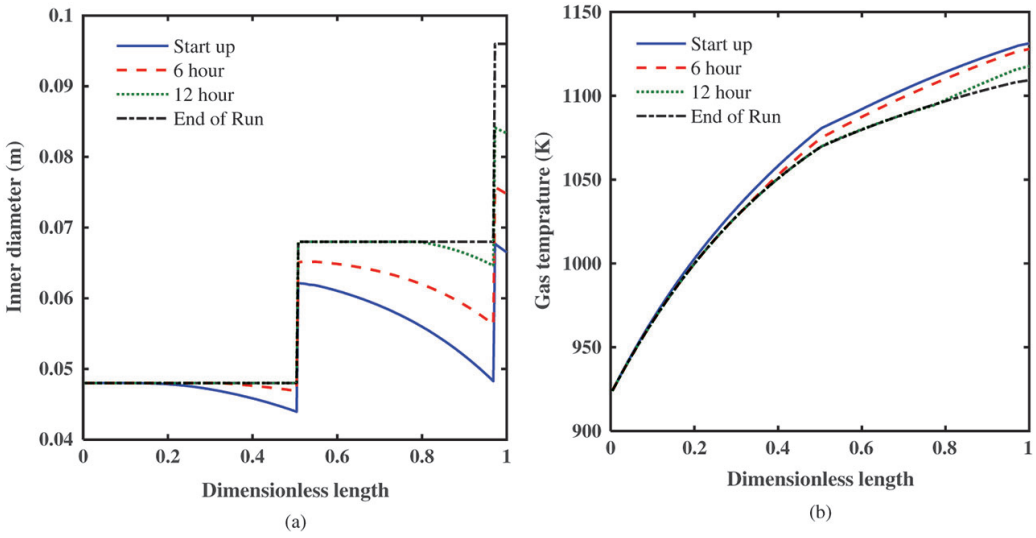


Figure 7: (a), Coke thickness versus tube length during the decoking process. (b), Gas phase temperature versus tube length during the decoking process.

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

5.4 Process optimization

In this section, the results of formulated optimization problem is presented at fixed run time. indeed the process run time is fixed at 60 days. Based on the considered constraints and bounds, feed temperature at initial and end of process run time are 888K and 905.3K, respectively. In addition, the furnace temperature should be increaesed gradually from 1435.5K to 1451.8K during the process run time. The optimum steam to naphtha ratio in the considered range is about 0.52. Applying the obtained optimal operating points increases ethylene production capacity from 153.7 mole s^{-1} to 155.6 mole s^{-1}. It appeared that ethylene production capacity is improved about 1.2%.

6 Conclusion

In this research, a mathematical pseudo-dynamic model was developed to simulate and optimization of naphtha cracking cycle based on the mass and energy balance equations. Comparison between simulation results and plant data proved the accuracy of developed model and assumptions. The results showed that coke deposition on the surface decreases rate of naphtha cracking considerably, so the ethylene production decreases from 155.5 $mols^{-1}$ at plant start up to 151.9 $mols^{-1}$ at the end of run time. In addition, the pressure drop at the start up and end of run times are 0.5 bar and 0.9 bar, respectively. It was appeared that the deposited coke on the downward tubes is burned after 8 h, while about 24 h is required to burn the deposited coke on the inner surface of upward tubes. Based on the formulated optimization problem to achieve maximum ethylene capacity, feed and furnace temperature should be increased from 888K to 905.3K, and 1435.5K to 1451.8K, respectively. The optimum steam to naphtha ratio in the considered range was about 0.52. The simulation results showed that, applying calculated optimal operating points on the system increases ethylene production capacity from 153.7 $mol.s^{-1}$ to 155.6 $mol.s^{-1}$.

Nomenclature

C concentration of coke ($molm^{-3}$)
C_p heat capacity ($Jmol^{-1}\ K^{-1}$)
d_t coil diameter (m)
F_i molar flow rate of component I ($moles^{-1}$)
F_r friction factor
G total mass flow rate ($Kgm^{-2}\ s^{-1}$)
h_i tube heat transfer coefficient($Wm^{-2}K^{-1}$)
h_f furnace heat transfer coefficient ($Wm^{-2}K^{-1}$)
k_i rate coefficient of reaction i (s^{-1}or $m^3mol^{-1}s^{-1}$)
M_c coke molecular weight ($Kgmol^{-1}$)
M_m average molecular weight ($Kgmol^{-1}$)
P_t total pressure (Pa)
Q heat flux (Wm^{-2})
R_c rate of coking ($mol\ m^{-2}\ s^{-1}$)
r_i rate of reaction i ($molm^{-2}\ s^{-1}$)
R universal gas constant ($Jmol^{-1}\ K^{-1}$)
R_i thermal resistance (KW^{-1})
R_b radius of the tube bend (m)
Re Reynolds number
S_{ij} stoichiometric coefficient of component j in ith reaction
T temperature (K)
t time (s)

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM

t_c coke thickness (m)

U overall heat transfer coefficient ($Jmol^{-1}\ K^{-1}$)

Z axial reactor coordinate (m)

Greek letters

ΔH heat of reaction ($KJmol^{-1}$)

ξ parameter of tube bend

Λ angel of bend

α coking factor

ρ_c coke specific gravity (Kgm^{-3})

References

[1] Yoshimura Y, Kijima N, Hayakawa T, Murata K, Suzuki K, Mizukami F, et al. Catalytic cracking of naphtha to light olefins. Catal Surv Jpn. 2001;4(2):157–167.
[2] Reyniers GC, Froment GF, F-D K, Zimmermann G. Coke formation in the thermal cracking of hydrocarbons. 4. Modeling of coke formation in naphtha cracking. Ind Eng Chem Res. 1994;33(11):2584–2590.
[3] Pant KK, Kunzru D. Pyrolysis of methylcyclohexane: Kinetics and modelling. Chem Eng J. 1997;67(2):123–129.
[4] Towfighi J, Niaei A, Karimzadeh R, Saedi G. Systematics and modelling representations of LPG thermal cracking for olefin production. Korean J Chem Eng. 2006;23(1):8–16.
[5] Ghassabzadeh H, Darian JT, Zaheri P. Experimental study and kinetic modeling of kerosene thermal cracking. J Anal Appl Pyrolysis. 2009;86(1):221–232.
[6] Belohlav Z, Zamostny P, Herink T. The kinetic model of thermal cracking for olefins production. Chem Eng Process Intensif. 2003;42(6):461–473.
[7] Gao GY, Wang M, Ramshaw C, Li XG, Yeung H. Optimal operation of tubular reactors for naphtha cracking by numerical simulation. Asia-Pacific J Chem Eng. 2009;4(6):885–892.
[8] Ghashghaee M, Karimzadeh R. Dynamic modeling and simulation of steam cracking furnaces. Chem Eng Technol. 2007;30(7):835–843.
[9] Keyvanloo K, Towfighi J, Sadrameli SM, Mohamadalizadeh A. Investigating the effect of key factors, their interactions and optimization of naphtha steam cracking by statistical design of experiments. J Anal Appl Pyrolysis. 2010;87(2):224–230.
[10] Keyvanloo K, Sedighi M, Towfighi J. Genetic algorithm model development for prediction of main products in thermal cracking of naphtha: Comparison with kinetic modeling. Chem Eng J. 2012;209:255–262.
[11] Barazandeh K, Dehghani O, Hamidi M, Aryafard E, Rahimpour MR. Investigation of coil outlet temperature effect on the performance of naphtha cracking furnace. Chem Eng Res Des. 2015;94:307–316.
[12] Kumar P, Kunzru D. Kinetics of coke deposition in naphtha pyrolysis. Can J Chem Eng. 1985;63(4):598–604.
[13] Kumar P, Kunzru D. Modeling of naphtha pyrolysis. Ind Eng Chem Des Dev. 1985;24(3):774–782.
[14] Zheng S, Zhang X, Qi C, Zhou H. Modeling of heat transfer and pyrolysis reactions in ethylene cracking furnace based on 3-D combustion monitoring. Int J Thermal Sci. 2015;94:28–36.
[15] Berreni M, Wang M. Modelling and dynamic optimization of thermal cracking of propane for ethylene manufacturing. Comput Chem Eng. 2011;35(12):2876–2885.
[16] Seifzadeh Haghighi S, Rahimpour MR, Raeissi S, Dehghani O. Investigation of ethylene production in naphtha thermal cracking plant in presence of steam and carbon dioxide. Chem Eng J. 2013;228:1158–1167.
[17] Cai H, Krzywicki A, Oballa MC. Coke formation in steam crackers for ethylene production. Chem Eng Process Intensif. 2002;41(3):199–214.
[18] Mitchell M. An introduction to genetic algorithms. Cambridge, MA: MIT press, 1998.
[19] Haupt RL, Haupt SE. Practical genetic algorithms. New York: John Wiley & Sons, 2004.

Automatically generated rough PDF by *ProofCheck* from River Valley Technologies Ltd.

Brought to you by | University of Gothenburg
Authenticated
Download Date | 10/3/17 9:24 AM